Does phylogenetic diversity predict ecosystem functioning in multi-trophic food webs?

Proposed framing — reanalysis of the T98 and T99 pond experiments

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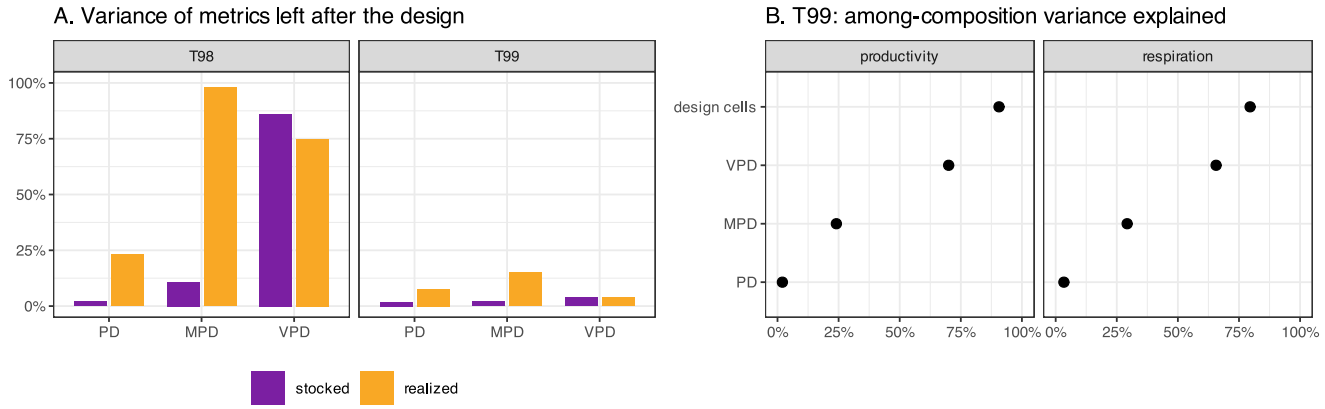
The question. A decade ago it was often argued that phylogenetic diversity (PD) should predict ecosystem function better than species richness, "in a range of communities, from single-trophic to complex networks" (Srivastava et al. 2012, *Ecol. Lett.*). The multitrophic part of that claim has rarely, if ever, been tested with experiments that manipulated more than two trophic levels, which are rare (e.g. of 94 diversity manipulations synthesized by Lefcheck et al. 2015, only 3 involved carnivores, each manipulating a single trophic group).

Why these data. The T98 (Downing & Leibold 2002, *Nature*) and T99 (Downing 2005, *Ecology*) experiments manipulated richness and composition of macrophytes, grazers and predators simultaneously, measured productivity, respiration and decomposition repeatedly, and imposed an acid pulse. They are among the very few designs able to test the claim directly.

Main results (Bayesian mixed models; details in the attached report).

1. **Whole-community phylogenetic metrics mostly restate trophic structure.** Every composition is built from plants and animals, and the plant–animal split dominates phylogenetic distances. Once the design is accounted for, little independent variation remains in the metrics of the stocked communities: 2% (PD) and 11% (MPD) in T98, and 2%–4% in T99 (panel A; VPD keeps more in T98, and realized compositions vary more because of extinctions and invasions).
2. **No robust effect of PD, MPD or VPD on productivity or respiration** at fixed richness or trophic design, for stocked or realized composition, in either experiment.
3. **Across trophic configurations (T99), one metric is an imperfect summary of food-web structure:** design cells (FG composition × richness) account for 91% / 80% of among-composition variance in productivity / respiration; the best single metric (VPD) for 70% / 66%; PD for ~0 (panel B).
4. **Disturbance:** the acid pulse depressed respiration by ≈ 1 SD, followed by recovery (stronger at the highest richness), consistent with Downing & Leibold (2010). Phylogenetic metrics did not robustly modulate resistance or recovery; a positive PD/MPD × recovery effect on decomposition in T99 was not replicated in T98 (exploratory).



Proposed message. In multitrophic communities, whole-community phylogenetic diversity largely re-describes who eats whom; once trophic structure and richness are controlled, it adds little. A phylogenetic "BEF shortcut" for food webs therefore requires designs in which relatedness varies within trophic levels, independently of trophic structure.

Proposed target: *Oikos* (Research article); fallback *Freshwater Biology*. **Working title:** "Phylogenetic diversity restates trophic structure and does not predict ecosystem functioning in experimental multitrophic food webs".

Questions for you: (i) Does this framing work for you? (ii) Amy: please confirm the timing of the acid pulse in T99 (inferred between days 19 and 22) and that the stocked compositions match your design. (iii) Any data or analyses you would add or remove?